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Agenda:

- Talk about tech
- Solve some pressing issues
- Live demo (hopefully)



Unusual uses of **Ansible Automation Platform** and **Event-Driven Ansible**

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Who am I?

- Rune Juhl
- Architect, Open Source, Atea Denmark
- Linux user for > 22 years
- Rabid advocate of open-source software
- Emacs user



Link to "Your pacemaker should be running open source software", a column in The Register by Steven J. Vaughan-Nichols

<https://www.theregister.com/2024/01/12/column/>

Who're we?

- Atea Denmark has a department called Open Source
- We work with... open source!
 - Linux! (all flavors)
 - Kubernetes, OpenShift, Rancher, Docker, Podman
 - Ansible, Puppet, Salt
 - DevOps, GitOps, SecOps, OpsOps
 - Prometheus, Thanos, Nagios, Zabbix

What is this about?

- Why play around with Ansible and EDA?
 - It's ~~fun~~ educational!
 - Exploring the boundaries
 - Determine what it's good at
 - Find interesting bugs
 - ...because why not?
- Beware!
 - This is exploration, not guaranteed to work
 - Not polished
 - No AI

Ansible and EDA primer

Ansible

- Infrastructure as Code (IaC)
 - Provisioning
 - Configuration management
 - Application deployment
- Agent-less
- Works everywhere
- Tons of modules

Event-Driven Ansible

- Make Ansible react to events
 - Remediation
 - Ticket enrichment
 - Risk mitigation
- Rulebooks define event flows
- Executes Ansible Playbooks
- Not so many modules



Red Hat Ansible Automation Platform

Ansible

```
1 ---
2
3 - name: 'Add monitoring tag to network devices'
4   hosts: all
5   gather_facts: false
6   module_defaults:
7     ansible.netcommon.telnet:
8       user: '{{ ios_username }}'
9       password: '{{ ios_password }}'
10      login_prompt: 'Username: '
11      timeout: 120
12      prompts:
13        - '[>#]$_'
14        - ': $_'
15
16   tasks:
17   - when: |-
18       (interfaces | length) > 0
19       or
20       (all_interfaces | default(False))
21     block:
22     - name: 'Check default device exec level'
23       ansible.builtin.include_tasks:
24         file: '_ios_check_exec_level_privileged.yml'
25
26     - name: 'Get interface list if required'
27       when: |-
28         all_interfaces | default(False)
29       ansible.builtin.include_tasks:
30         file: '_ios_get_interfaces.yml'
31
```

```
1 --- Ansible Rulebook - Ansible rulebook files (ruleset_schema.json)
2 - name: http sse events
3   hosts: localhost
4   default_events_ttl: "2 seconds"
5   match_multiple_rules: false
6   sources:
7     - runejuhl.unusual_ansible.http_sse:
8       url: '{{ loo_roll_basename }}/events'
9       event_ids:
10         - binary_sensor-toilet_paper_situation_
11
12   rules:
13     - name: Detect no paper left on loo roll
14       condition:
15         all:
16           - event.data.value
17
18       actions:
19         - run_job_template:
20             name: loo_roll_empty
21             organization: Default
22
23         - run_job_template:
24             name: send_notification
25             organization: Default
26             job_args:
27               extra_vars:
28                 message: |-
29                   TP roll empty!
30                 icon: |-
31                   /usr/share/notify-osd/icons/Humanity/scalable/status/notification-printer-error.s
```


Ansible and EDA common use cases

Ansible

- Mass config change
 - Servers, desktops, hardware
- Automated provisioning
 - VMware, AWS/GCP/Azure, ++
- Security audit
 - OpenSCAP, ComplianceAsCode
- Deployment
 - Package manager, Kubernetes, Docker/Podman, AWS Lambda

Event-Driven Ansible

- Extend disks before full
- Change legacy system passwords
 - Keep in sync with AD/FreeIPA/etc.
- Augment incidents
 - Add logs, metrics, other details
- Respond to security events from EDR, NDR
 - E.g. quarantine device

Goal: do something unusual

Ansible and Event-Driven Ansible:

- Test platform flexibility
- Explore the boundaries
- Demonstrate Ansible as “glue”
- Try out multiple EDA event sources
- Work with hardware
 - ...because live demos are fun. Sometimes.

Why home automation?

- Long-time Home Assistant user
- "Programming" in YAML is not fun
- CLI >> GUI
- A chance to play with hardware
- ...because I felt like it!



Link to "Why the fuck are we templating yam!?", a blog post by Lee Briggs

<https://leebriggs.co.uk/blog/2019/02/07/why-are-we-templating-yaml>

GUI fatigue

Blueprint

Blueprint to use
Controller - IKEA E1743 TRÅDFRI On/Off Switch & Dimmer

Controller - IKEA E1743 TRÅDFRI On/Off Switch & Dimmer

Controller automation for executing any kind of action triggered by the provided IKEA E1743 TRÅDFRI On/Off Switch & Dimmer. Allows to optionally loop an action on a button long press.
Supports deCONZ, ZHA, Zigbee2MQTT.

Automations created with this blueprint can be connected with one or more [Hooks](#) supported by this controller.
Hooks allow to easily create controller-based automations for interacting with media players, lights, covers and more.
See the list of [Hooks available for this controller](#) for additional details.

Full documentation regarding this blueprint is available [here](#).

This blueprint is part of the [Awesome HA Blueprints](#) project.

Version 2023.10.05

(Required) Integration

Integration used for connecting the remote with Home Assistant. Select one of the available values.

☐ deCONZ

☐ ZHA

☒ Zigbee2MQTT

(deCONZ, ZHA) Controller Device

The controller device to use for the automation. Choose a value only if the remote is integrated with deCONZ, ZHA.

Device

(Zigbee2MQTT) Controller Entity

The action sensor of the controller to use for the automation. Choose a value only if the remote is integrated with Zigbee2MQTT.

Entity
ikea_e1743_switch_01 Action

(Required) Helper - Last Controller Event

Input Text used to store the last event fired by the blueprint. You will need to manually create a text input entity for this, please read the blueprint Additional Notes for more info.

Entity
ikea_e1743_switch_01_helper

(Optional) Up button short press

Action to run on short up button press.

Perform action 'Light: Turn on' on Bedroom ceiling lamp

+ ADD ACTION + ADD BUILDING BLOCK

(Optional) Up button long press

Action to run on long up button press.

+ ADD ACTION + ADD BUILDING BLOCK

(Optional) Up button release

Action to run on up button release after long press.

+ ADD ACTION + ADD BUILDING BLOCK

(Optional) Up button double press

Action to run on double up button press.

Light 'Turn on' on Bedroom

+ ADD ACTION + ADD BUILDING BLOCK

YAML fatigue

```
174 condition:
175   # if triggered by the homeassistant platform check if the automation should be run, as specified by the user
176   # if triggered by something else, just continue
177   - condition: and
178     conditions:
179       # if triggered by the run time, check that the automation is allowed to run today
180       - >-
181         {%- set today = now().weekday() -%} {{ trigger.id != "run_time" or today == 0 and day_monday or today == 1 and day_tuesday or
182         # if triggered due to home assistant startup, check that the feature is enabled
183         - '{{ trigger.id != "homeassistant_start" or trigger_at_homeassistant_startup }}'
184         # if triggered due to home assistant startup or to a custom trigger event, perform the execution check
185         # check whether the automation run at or after the expected run time
186         # if this is the case, there's no need to run the automation again
187         # else, the automation was not able to run at the expected time, and was not executed after it, hence run it now
188         # if block late execution is enabled, check also that no more than the provided max elapsed time has passed since the expected r
189         - >-
190         {%- set cdt = now() | as_timestamp | timestamp_local | as_datetime -%}
191         {%- set lsrdt = ((states(helper_storage) | from_json).last_triggered if (states(helper_storage) | regex_match(json_regex)) els
192         {%- set idt = strptime(states(time_entity),time_fmt) -%}
193         {%- set days = [day_monday, day_tuesday, day_wednesday, day_thursday, day_friday, day_saturday, day_sunday] -%}
194         {%- set cwd = cdt.weekday() -%}
195
196         {% macro day_offset(day) -%}
197           {%- if days[day] -%}
198             {{ 7 if cwd == day and cdt.time() < idt.time() else (cwd-day)%7 }}
199           {%- else -%}
200             -1
201           {%- endif -%}
202         {%- endmacro %}
203
204         {%- set weekdays = [day_offset(0)|int, day_offset(1)|int, day_offset(2)|int, day_offset(3)|int, day_offset(4)|int, day_offset(
205
206         {%- if weekdays | length > 0 -%}
207           {%- set closest_weekday = weekdays | min -%}
208           {%- set day_diff = timedelta(weeks=weeks_frequency-1, days=closest_weekday) -%}
209           {%- set edt = (cdt - day_diff - timedelta(hours=cdt.hour, minutes=cdt.minute, seconds = cdt.second) + timedelta(hours=idt.ho
210           {{ trigger.id not in ["homeassistant_start","custom_trigger_event"] or lsrdt < edt and (not block_late_execution or edt >= c
211         {%- else -%}
212           {{ false }}
213         {%- endif -%}
214
215 action:
216   # store the current datetime into the helper, since the automation was successfully triggered
217   - service: input_text.set_value
218     data:
```

Let's replace it!

- ...with Ansible and more YAML!
- Increased flexibility?
 - Use any programming languages (execution/decision environments)
 - Lots of Ansible modules available
- How hard can it be?
- ...he said, before spending countless hours debugging and monkey-patching EDA Decision Environments



Home automation with Ansible/EDA

Trigger sources

- MQTT
 - Zigbee2MQTT
 - DNS server logs
- HTTP server-sent events (SSE)
 - ESPHome devices
- Websocket
 - Home Assistant
- Polling HTTP
 - Scraping

Respond using Ansible

- MQTT
 - Home Assistant
- HTTP REST API
 - UniFi Network Application (undocumented?)
 - Home Assistant (slightly less undocumented)
 - Robot Bartender
- UDP?
- Anything you can think of

Ideas



Listen to presence sensor, turn on lights



Throttle Wi-Fi if the kids play Roblox when they should sleep



Detect no more toilet paper



Robot Bartender: have it serve drinks when you get home



Use mm-Wave radar, control smoke machine for scary ambiance



Use Zigbee remote to control computer

Throttling the kids

Imagine the following

- Kids play Roblox/Fortnite/Among Us
- They become addicted
 - Drops out of school
 - Become game developers

Solution

- Automatically throttle Wi-Fi access using EDA!

Concept

- Pi-Hole has DNS logs
- Roblox talks to domains that match `.*roblox\.com$`
- Logstash can ingest and filter
- Push to MQTT for events
- EDA listens to events, does stuff



EDA and MQTT

- EDA uses event source plugins
 - ...MQTT not supported by default
- The power of open source
 - Thank you, Colin McNaughton!
- Only needed a few library updates to work

The screenshot shows the GitHub interface for the repository 'cloin / event-driven-minecraft'. The 'Files' tab is active, showing the directory structure. The 'mqtt.py' file is selected in the 'eda' directory. The code editor displays the content of 'mqtt.py', which is an Ansible rulebook plugin for receiving events via MQTT. The code includes arguments for host and topic, test script instructions, and an Ansible rulebook configuration with a name 'New minecraft event', a condition 'event.type is defined', and an action 'debug'.

```
1 """
2 mqtt.py
3
4 An ansible-rulebook event source plugin for receiving events via
5
6 Arguments:
7     host:      The host where the mqtt topic is hosted
8     topic:     The mqtt topic
9
10 Test script stand-alone by exporting MQTT_HOST and MQTT_TOPIC as
11 - $ export MQTT_HOST=localhost
12 - $ export MQTT_TOPIC=messages
13
14 In an EDA rulebook (note: fake collection, cloin.minecraft does
15
16 - name: Minecraft events
17   hosts: localhost
18   sources:
19     - cloin.minecraft.mqtt:
20       host: localhost
21       topic: messages
22
23   rules:
24     - name: New minecraft event
25       condition: event.type is defined
26       action:
27         debug:
28
29 """
30
31 import asyncio
32 import logging
33 import os
34 from typing import Any, Dict
35
36 from asyncio_mqtt import Client
```

Pipeline: Logs

```
Oct 28 17:19:49 dnsmasq[198080]: query[A] clientsettingscdn.roblox.com from 192.168.1.226
Oct 28 17:19:49 dnsmasq[198080]: forwarded clientsettingscdn.roblox.com to 208.67.222.222
Oct 28 17:19:49 dnsmasq[198080]: reply clientsettingscdn.roblox.com is <CNAME>
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.84
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.94
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.16
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.77
```

Pipeline: Logstash

```
input {  
  file {  
    path => ['/var/log/pihole.log']  
    mode => 'tail'  
    start_position => 'end'  
  }  
}  
filter {  
  mutate { gsub => ["message", "^.+?: ", ""] }  
  if ([message] !~ "^query") { drop {} }  
  dissect {  
    mapping => { message => "query[%{type}] %{name} from %{source}" }  
  }  
  prune { whitelist_names => ["^type$", "^name$", "^source$"] }  
  if ([source] =~ "^(127\.|172\.)") { drop {} }  
}  
output {  
  stdout { codec => 'json' }  
  mqtt {  
    host => "192.168.1.243"  
    port => 1883  
    topic => "dns"  
  }  
}
```


Pipeline: EDA

```
- name: mqtt events
hosts: localhost
sources:
  - runejuhl.unusual_ansible.mqtt:
      host: '{{ dns_mqtt_server }}'
      topic: 'dns'

rules:
  - name: Run job on DNS lookup
    # match hours 21, 22, 23, 00, 01, 02, 03, 04, 05, 06, 07 OR the right
    # dates
    condition: |-
      event.payload.name is regex('roblox\.com$')
      and
      (
        event.meta.received_at is regex('T(2[0-3]|0[0-7]):')
        or
        event.meta.received_at is regex('2024-11-28T')
        or
        event.meta.received_at is regex('2024-12-05T')
      )
    actions:
      - run_job_template:
          name: send_notification
          organization: Default
          job_args:
            extra_vars:
              message: |-
                IP {{ event.payload.source }} has been naughty!
              icon: |-
                /usr/share/notify-osd/icons/Humanity/scalable/status/notifica
      - run_job_template:
          name: unifi_throttle
          organization: Default
          job_args:
            extra_vars:
              throttle_client: true
```

- No Ansible module (yet)
- Get CSRF token from API
- Login with username/password
- Get list of active devices
- Find device with matching IP
- Call API and throttle device

```
name: |-
  Throttle wifi
hosts: localhost
gather_facts: false
tasks:

- name: Login
  ansible.builtin.uri:
    url: '{{ unifi_baseurl }}/api/login'
    method: POST
    headers:
      content-type: 'application/json'
    return_content: true
    body:
      {
        "username": "{{ unifi_username }}",
        "password": "{{ unifi_password }}",
        "remember": false,
        "strict": true
      }
    body_format: 'json'
  register: login

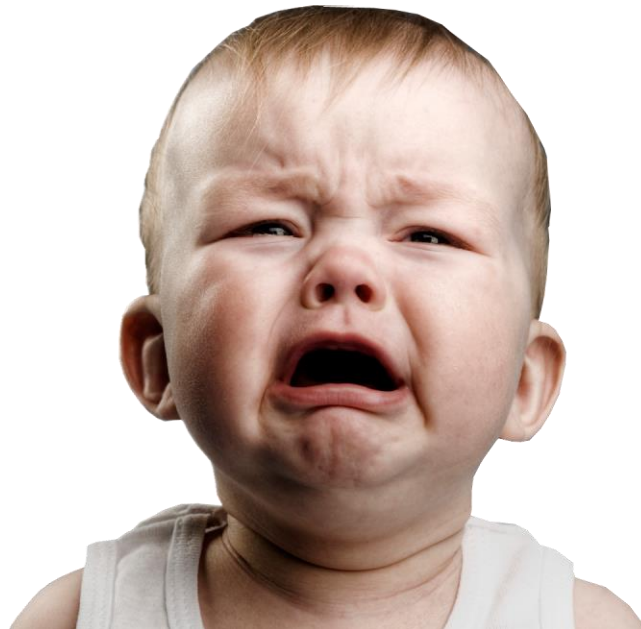
- name: Get devices
  ansible.builtin.uri:
    url: '{{ unifi_baseurl }}/v2/api/site/default/clients/active?includeTrafficUsage=true&includeUnifiDevices=true'
    headers:
      Cookie: |-
        {{ login.cookies_string }}
      X-Csrf-Token: |-
        {{ login.cookies.csrf_token }}
    return_content: true
  register: devices

- ansible.builtin.set_fact:
  device: |-
    {{ devices.json|selectattr('ip', 'eq', ansible_edat.event.payload.source)|first }}

- name: Throttle device
  ansible.builtin.uri:
    url: '{{ unifi_baseurl }}/api/s/default/rest/user/{{ device.user_id }}'
    method: PUT
    headers:
      Cookie: |-
        {{ login.cookies_string }}
      X-Csrf-Token: |-
        {{ login.cookies.csrf_token }}
    return_content: true
    body:
      usergroup_id: |-
        {% if throttle_client %}5ff6a0c246e0fb01138ed94f{% endif %}
    body_format: 'json'
  register: call_throttle

- ansible.builtin.debug:
  msg: |
    {{ call_throttle.json }}
  when: |-
    not call_throttle.json.meta.rc == "ok"
```

Job done!



Kid #1 tablet



Name

Kid #1 tablet

Note

WiFi Speed Limit ⓘ

Tarpit

☐ Lock to Access Point ⓘ

IP Settings

☐ Fixed IP Address

Reconnect

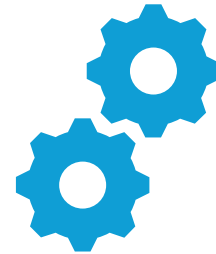
Empty TP roll detector



Common household
issue



Not fun being left out to
dry



A case for automation!

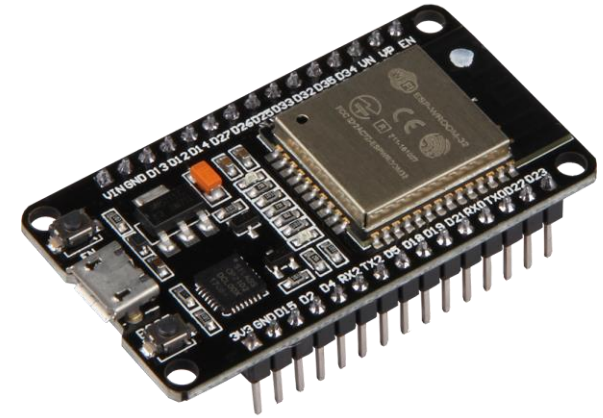
Concept phase

- TP roll state $\approx$ color of roll
 - High color temperature: good
 - Low color temperature: bad
- Detect color using LED?
 - ...or see what they have on eBay
- Use Server-Sent Events
- Trigger Home Assistant using EDA



Technical specs

- NodeMCU dev board (ESP8266)
- TCS34725 I²C RGB sensor
 - I²C is nice: 2 wires + power
- Li-Ion 18650 battery holder
- ESPHome
 - Even more YAML!



ESPHome

- Make IoT devices in YAML
 - ...complete with in-line C++
- Decent documentation
- Huge community
- And lots of YAML!

```
sensor:
- platform: tcs34725
  glass_attenuation_factor: 1.0
  address: 0x29
  update_interval: 100ms
  color_temperature:
    name: "TCS34725 Color Temperature"
    internal: true
  filters:
    - sliding_window_moving_average:
        window_size: 100
        send_every: 25
        send_first_at: 25
    - lambda: |-
        static bool initial_run = true;
        static bool problem = false;

        if (initial_run) {
            ESP_LOGE("main", "setting initial state: %d", problem);
            initial_run = false;
            id(toilet_paper).publish_state(problem);
            return x;
        }

        if (x >= 5000) {
            ESP_LOGE("main", "got crazy color temperature: %f", x);
            return {};
        }

        if (x < 3100.0) {
            if (!problem) {
                ESP_LOGW("main", "Houston, we have a problem! Color
```

Server-Sent Events

- Comes with ESPHome
- Very simple protocol
 - HTTP server just keeps sending data
- No SSE event source plugin
 - Made one based on MQTT plugin
 - Added matching on event IDs

```
- name: http sse events
  hosts: localhost
  default_events_ttl: "2 seconds"
  match_multiple_rules: false
  sources:
    - http_sse:
        url: '{{ loo_roll_basename }}/events'
        event_ids:
          - binary_sensor-toilet_paper_situation_

  rules:
    - name: Detect no paper left on loo roll
      condition:
        all:
          - event.data.value

  actions:
    - run_job_template:
        name: loo_roll_empty
        organization: Default
```

Home Assistant • now 



Loo roll empty, again!

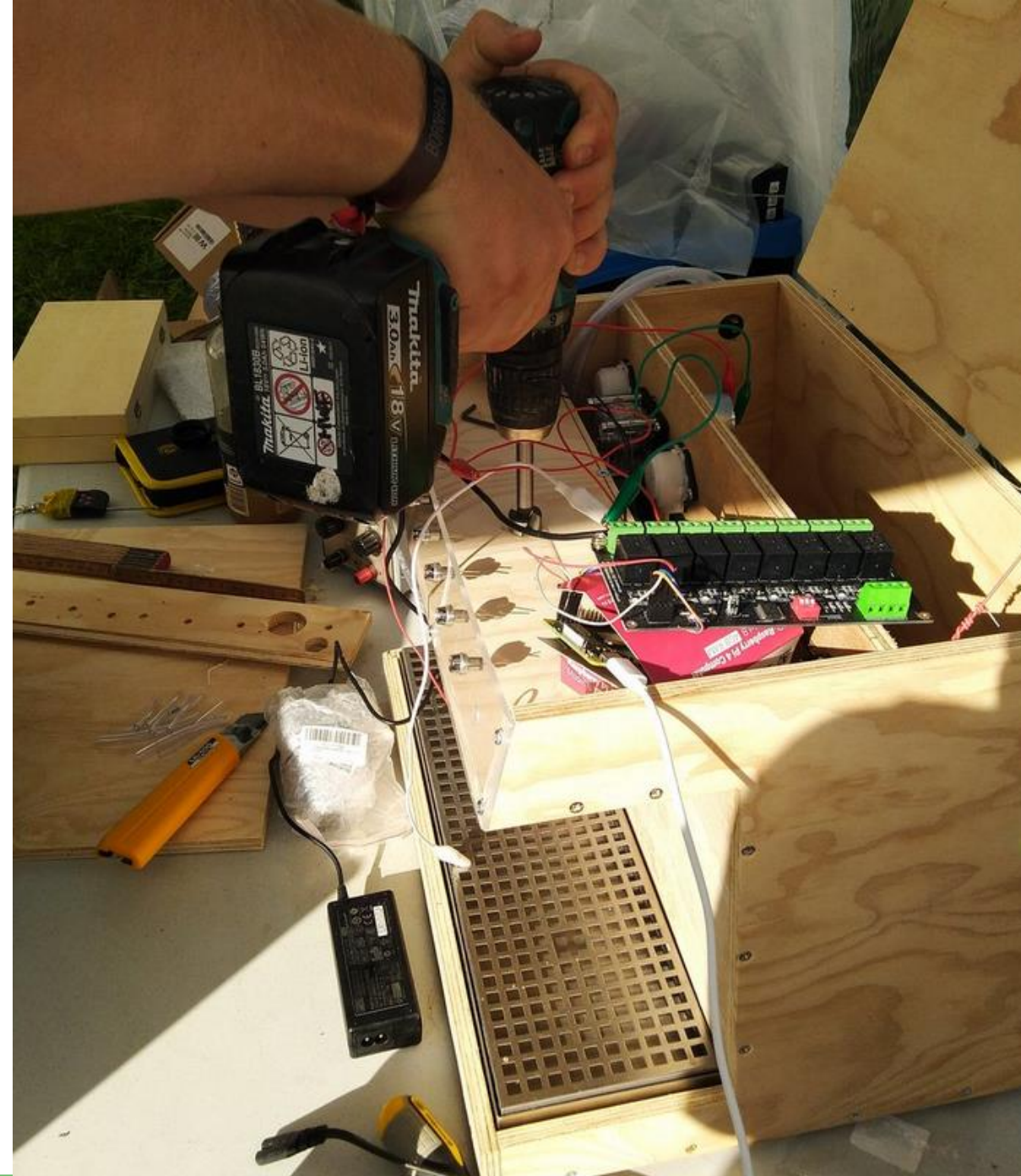
Whodunnit?



Future improvements?

- ESP32-based BLE presence detector?
 - Bluetooth MACs aren't stable (privacy)
- Facial recognition using Frigate and Double Take?
- ...maybe an accessible TP roll stock is easier...

Robot Bartender



Robot Bartender

Built as a project for BornHack 2021

- Raspberry Pi
- I²C relays
- 10 peristaltic pumps
- Backend in Clojure
- Web frontend in PowerShell



Robot Bartender: State

Enterprise
ready!



Almost worked at the time



In storage ever since

...much to the
dismay of my kids
Until now!



Fixed the last bug (I hope)



Simple HTTP API (no security)

Robot Bartender: Goal

1. Listen for presence events
2. Make drinks automatically
3. ...?
4. Profit!

Bar menu

- Job templates give us a GUI
- Menu implemented as a "Survey"
 - Not dynamic (but there's an open issue on GitHub...)
- Template has access controls
 - Kids only allowed virgin cocktails
- Audit logs are built-in
 - Easily check alcohol consumption

Templates ?

A job template is a definition and set of paramet

☐ ▾

▾

▾

Name ↑

- > ☐ Make adult drink
- > ☐ Make virgin drink

[Templates](#) > [Make virgin drink](#)

Prompt on Launch

1 Survey

2 Review

Which drink? *

Select option

virgin-bloody-mary

virgin-pina-colada

orange-juice

exotic-fruits-juice

apple-juice

Event-Driven Drinks

- Home Assistant knows when I'm home
 - Automatic drinks!
- EDA `url_check` can poll HTTP
 - ...but no authentication
- Fork and patch
 - Minor changes to allow authorization header

```
- name: 'Drinks by Home Assistant state'
hosts: localhost
sources:
  - url_probe:
      urls:
        - '{{ hass_baseurl }}/api/states/person.rune'
      headers:
        Authorization: 'Bearer {{ hass_token }}'
        content-type: application/json
      delay: 5

rules:
  - name: Any event
    condition:
      all:
        - event.meta.source.name == 'url_probe'
        - event.url_probe.body.state == 'not_home'
        - facts.last_state != 'not_home'

    actions:
      - run_job_template:
          name: make_drinks
          organization: Default
          job_args:
            extra_vars:
              drink_name: gt
```

Home Automation using Ansible?

Pros

- Everything worked!
- Very flexible
- Good auditing capabilities
- EE/DE is nice to have
- Enterprise support?

Cons

- Small community
- Resource hog + slow
- Ansible Rulebooks != EDA
- Family didn't like the GUI

That's it!



Robot Bartender worked?
Come and have a drink!



<https://github.com/runejuhl/unusual-ansible>
https://github.com/runejuhl/runejuhl.unusual_ansible
<https://github.com/runejuhl/esphome-loo-roll-detector>